Safety update: Tepotinib in *MET*ex14 skipping NSCLC

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CONCLUSIONS

- **In the VISION study, comprising a large population of patients with *MET*ex14 skipping NSCLC, tepotinib was generally well tolerated**
- **The most common TRAE, peripheral edema, which is a class effect of MET inhibitors, was mostly mild or moderate and managed with tepotinib dose reduction or treatment interruption**
- **Proactive monitoring for edema is recommended, with consideration of early dose modifications (e.g. dose reductions or short treatment interruptions) or prophylactic conservative management measures, such as support stockings, limb elevation, and increased physical activity**

VISION study design

- VISION is a single-arm, Phase II trial of the MET inhibitor tepotinib in patients with NSCLC harboring *MET* alterations (NCT02864992)[1]
- Based on results from the Phase II VISION study, tepotinib was approved for advanced NSCLC with *MET*ex14 skipping in the US in February 2021[2]

Figure 1. Study design, endpoints, and eligibility criteria of VISION

*Treatment continues until disease progression, intolerable toxicity, or withdrawal of consent.

Selected endpoints	
Primary	•Objective response by IRC according to RECIST v1.1 criteria
Secondary	•DOR, PFS •Safety – TRAEs were assessed using NCI-CTCAE v4.03 – In-range AEs, treatment could be interrupted for up to 21 days or the tepotinib dose could be reduced to 250 mg*

*Reduced 250 mg dose contains 225 mg tepotinib free base.

Key inclusion criteria
- Advanced NSCLC (*EGFR*/*ALK* wild-type, all histologies)
- Liquid and/or tissue biopsy MET alterations (central lab)
- First, second or third line of therapy
- Prior immunotherapy allowed

Key exclusion criteria
- Any unresolved toxicity Grade 2 or more from previous anticancer therapy
- Inadequate hematologic, liver or renal function, impaired cardiac function, uncontrolled hypertension

Safety analysis: Methods

- Treatment-emergent AEs were graded according to NCI-CTCAE v4.03

Treatment-related AEs were defined as
- **Absent** pretreatment
- **Worsened** relative to pretreatment, with onset occurring between the first dose and 30 days after the last dose

Investigators assessed the causal relation to tepotinib based on factors including
- Safety profile
- Medical history
- Concomitant medication
- Course of the underlying disease
- Trial procedures

- Time to first onset and time to resolution were analyzed for AEs of clinical interest, including composite categories comprising preferred terms, irrespective of the causal relation to study treatment

Peripheral edema – a disorder characterized by swelling due to excessive fluid accumulation in the upper or lower extremities
NCI CTCAE v4.03 definitions

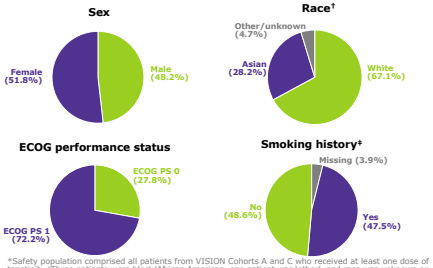
Grade 1	5–10% inter-limb discrepancy in volume or circumference at point of greatest visible difference; swelling or obscuration of anatomic architecture on close inspection
Grade 2	>10–30% inter-limb discrepancy in volume or circumference at point of greatest visible difference; readily apparent obscuration of anatomic architecture; obliteration of skin folds; readily apparent deviation from normal anatomic contour; limiting instrumental ADL
Grade 3	>30% inter-limb discrepancy in volume; gross deviation from normal anatomic contour; limiting self-care ADL

RESULTS

VISION: Patient characteristics and efficacy

- Patients with *MET*ex14 skipping were elderly (median age 72.0 years, range: 41–94), approximately half were male, and half had a history of smoking

Baseline characteristics (N=255)*



*Safety population comprised all patients from VISION Cohorts A and C who received at least one dose of tepotinib. †Three patients were black/African American, one patient was 'other', and race was unknown or missing in eight patients. ‡Smoking history was missing in ten patients.

- Tepotinib had robust and durable efficacy[1,3]

Efficacy (IRC-assessed)

Efficacy according to IRC	N=152*
ORR, % (95% CI)	**44.7** (36.7, 53.0)
Median DOR, months (95% CI)	**11.1** (8.4, 18.5)
Median PFS, months (95% CI)	**8.9** (8.2, 11.2)

*The efficacy analysis population comprised all patients from VISION Cohort A (data cut-off: July 1, 2020).

Overall safety profile of tepotinib

- Median treatment duration in the safety population was 5.1 months (range <1–43.3) with treatment still ongoing in 101 patients (39.6%)
- Most treatment-related AEs were mild-to-moderate

Tepotinib safety profile

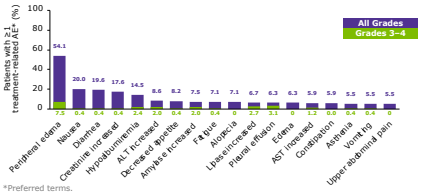
AEs, n (%)	*MET*ex14 skipping N=255*	
	All cause	Treatment-related
All grades	246 (96.5)	220 (86.3)
Serious AEs	115 (45.1)	31 (12.2)
Grade ≥3	135 (52.9)	64 (25.1)
Leading to dose reduction	76 (29.8)	71 (27.8)
Leading to treatment interruption	112 (43.9)	90 (35.3)
Leading to permanent discontinuation	52 (20.4)	27 (10.6)

*Safety population comprised all patients from VISION Cohorts A and C who received at least one dose of tepotinib (data cut-off: July 1, 2020).

Treatment-related AEs

- Peripheral edema was the most common treatment-related AE
- ILD-like events occurred in six patients (2.4%)

Treatment-related AEs (any Grade) occurring in ≥5% of patients (N=255)



*Preferred terms.

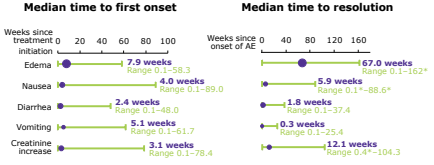
- Three patients had fatal treatment-related AEs
  - Acute respiratory failure secondary to ILD (n=1)
  - Severe worsening of dyspnea (n=1)
  - Acute hepatic failure after they had withdrawn consent (n=1)

AEs of clinical interest

AEs (all cause) of clinical interest, n (%)	Overall N=255		
	All Grades	Grade 3	Grade 4
Edema (composite term)	**178 (69.8)**	**24 (9.4)**	**0**
Peripheral edema	153 (60.0)	20 (7.8)	0
Edema	18 (7.1)	0	0
Generalized edema	13 (5.1)	5 (2.0)	0
Face edema	7 (2.7)	0	0
Localized edema	6 (2.4)	1 (0.4)	0
Genital edema	6 (2.4)	3 (1.2)	0
Periorbital edema	2 (0.8)	0	0
Scrotal edema	2 (0.8)	0	0
Peripheral swelling	1 (0.4)	0	0
Gastrointestinal AEs of clinical interest			
Nausea	68 (26.7)	2 (0.8)	0
Diarrhea	67 (26.3)	1 (0.4)	0
Vomiting	33 (12.9)	3 (1.2)	0
Creatinine increase (composite)	**66 (25.9)**	**1 (0.4)**	**0**
Blood creatinine increased	64 (25.1)	1 (0.4)	0
Hypercreatininemia	2 (0.8)	0	0

Time to first onset and time to resolution

- Median time to first onset of gastrointestinal AEs (2.4–5.1 weeks) and creatinine increase (3.1 weeks) were shorter than edema (7.9 weeks)



AE (patients with ≥1 event)	Edema (n=178)	Nausea (n=68)	Diarrhea (n=67)	Vomiting (n=33)	Creatinine increase (n=66)
Total events	337	87	112	47	96
Events resolved at time of analysis	115	67	102	44	67

Analyses of time to first onset and time to resolution were carried out for AEs of clinical interest, including composite categories comprising preferred terms, and were analyzed irrespective of causal relation to study treatment. Time to first onset was described by median and range for observed AEs, not accounting for competing events. Time to resolution was analyzed using Kaplan–Meier methods in a descriptive manner, not accounting for the fact that one patient could contribute by more than one event of the respective AE. *Denotes a censored value.

Management of AEs of clinical interest

Tepotinib dosing recommendations for AEs of clinical interest presented:[2]

Grade 2: Maintain dose level. If intolerable, consider withholding tepotinib until resolved, then resume at a reduced dose

Grade 3: Withhold tepotinib until resolved, then resume at a reduced dose

Grade 4: Permanently discontinue tepotinib

Management of edema (composite term)

- Edema was manageable, and resulted in few permanent treatment discontinuations

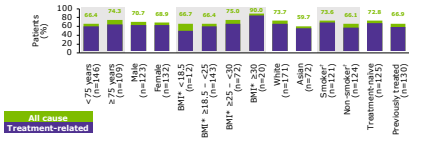
Management recommendations
- Proactive monitoring for edema, including weight measurement and peripheral measurement of limbs at baseline to detect increases
- If weight/peripheral circumference increase, initiate management measures (e.g. support stockings, limb elevation, increased physical activity, kinesiotherapy)
  - Cross-functional management in a lymphedema clinic can also be considered[4]
- Dose modifications such as dose reduction or temporary treatment interruption should be considered early to mitigate severity of edema

Frequency of treatment modifications due to edema

Patients with at least one event leading to:	Edema (all cause)
Dose reduction, n (%)	48 (18.8)
Temporary interruption, n (%)	59 (23.1)
Permanent discontinuation, n (%)	11 (4.3)

Management of edema (composite term) contd.

Edema incidence across clinically relevant subgroups



*BMI was missing for eight patients; †Smoking history was missing for ten patients.

Management of nausea, diarrhea and vomiting

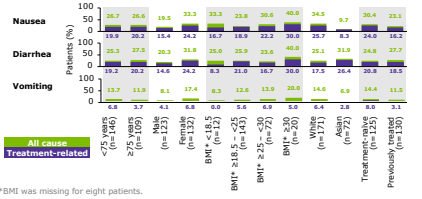
- A low proportion of patients had a treatment modification as a result of gastrointestinal AEs

Management recommendations
- Tepotinib should be taken soon after a meal, which may decrease the risk of gastrointestinal AEs
- Diarrhea can be managed with standard anti-diarrheal treatments such as loperamide, and treatment can be temporarily interrupted to manage gastrointestinal AEs

Frequency of treatment modifications due to gastrointestinal AEs

Patients with at least one event leading to:	Nausea (all cause)	Diarrhea (all cause)	Vomiting (all cause)
Dose reduction, n (%)	2 (0.8)	0	0
Temporary interruption, n (%)	5 (2.0)	5 (2.0)	1 (0.4)
Permanent discontinuation, n (%)	1 (0.4)	1 (0.4)	0

Incidence of gastrointestinal AEs across clinically relevant subgroups



*BMI was missing for eight patients.

Management of creatinine increase (composite term)

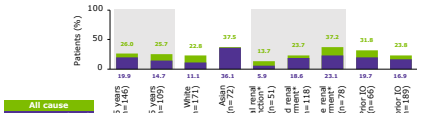
- Treatment modifications as a result of creatinine increase were uncommon

Management recommendations
- Patients typically experience a rapid creatinine increase that plateaus without renal impairment
  - Based on non-clinical studies, increases in creatinine may reflect direct inhibitory effects on renal tubular transporters[5–7]
- Alternative markers for measuring glomerular filtration rate can determine if creatinine elevation reflects renal impairment

Frequency of treatment modifications due to creatinine increase

Patients with at least one event leading to:	Creatinine increase (all cause)
Dose reduction, n (%)	7 (2.7)
Temporary interruption, n (%)	16 (6.3)
Permanent discontinuation, n (%)	2 (0.8)

Creatinine increase incidence across clinically relevant subgroups



*Renal impairment status was missing for eight patients.

Abbreviations: ADL: activities of daily living; AE: adverse event; ALK: anaplastic lymphoma kinase; CI, confidence interval; DOR, duration of response; ECOG PS, Eastern Cooperative Oncology Group performance status; EGFR, epidermal growth factor receptor; ILD, interstitial lung disease; IO, immuno-oncology therapy; IRC, independent review committee; LBx, liquid biopsy; MET, mesenchymal-epithelial transition factor; *METex14*, *MET* exon 14; NCI CTCAE, National Cancer Institute Common Terminology Criteria for Adverse Events; NSCLC, non-small cell lung cancer; ORR, objective response rate; PFS, progression-free survival; QD, once daily; RECIST, Response Evaluation Criteria in Solid Tumors; TBx, tissue biopsy; TRAE, treatment related adverse event.

References: 1. Paik PK, et al. *N Engl J Med*. 2020;383(10):931–43; 2. Tepotinib US Prescribing Information, 2021; 3. Paik PK, et al. Presented at WCLC 2021. Abstract MA11.05; 4. Goodwin K, et al. Presented at NACCL-2020. Abstract MO01.04; 5. Arakawa H, et al. *J Pharm Sci*. 2017;106(9):2890–2903; 6. Ongole S, et al. *Sci Rep*. 2018;8(1):9237; 7. Mohan A, et al. *Am J Kidney Dis*. 2021:S0272-6386(21)00640-5.

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